

# Prachatos Mitra

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## SUMMARY

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Computer architecture PhD candidate in performance modeling and bottleneck analysis for large-scale systems, with performance analysis and hypervisor systems work experience at AMD and Nutanix respectively. Seeking Summer 2027 performance modeling / architecture internships in the US.

## EDUCATION

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**Georgia Institute of Technology**

January 2023 - December 2027 (expected)

*PhD, Computer Science*

GPA: 4.0/4.0

Advisor: Dr. Alexandros Daglis

Research Area: Performance modeling and bottleneck analysis for large-scale systems; scale-out graph processing architectures

**Indian Institute of Science**

August 2018 - July 2020

*M.Tech., Computer Science*

GPA: 8.80/10.00

M.Tech thesis advisor: Dr. Arkaprava Basu

Research Area: Memory systems - TLB dead page and cache dead-block prediction

**Maulana Abul Kalam Azad University of Technology**

August 2014 - July 2018

*B.Tech., Computer Science and Engineering*

GPA: 9.01/10.00

## RESEARCH EXPERIENCE

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### • Optimizing resource allocation for actor-based workloads on commodity hardware

*Georgia Institute of Technology*

2026-Present

- Profiled performance characteristics of actor-based workloads on commodity hardware as a function of CPU frequency, LLC size and memory bandwidth
- Designing a runtime-level detector for imbalance of resources and work and dynamically adjusting resources

### • Design and simulation for actor-model based sparse graph processing architecture

*Georgia Institute of Technology*

2023-Present

- Helped design messaging and network-connected components for a novel scale-out actor-model graph processing architecture, under the IARPA AGILE program
- **MacroPattern:** abstracted actor-model workloads into six fitted distributions per handler, scaled analytically to arbitrary problem and system size via two-level curve fitting
- **MPSim:** SST-based parallel discrete-event simulator consuming scaled distributions; simulates ~16 million processing elements at ~8 MB per simulated core. 2.5–30× more accurate than regression-based scaling across graph microbenchmarks; workload representation seven orders of magnitude smaller than equivalent traces
- Exposed a head-of-line blocking pathology in the target architecture's node-level queueing, visible only at scale

### • Dead Page and Dead Block Predictors: Cleaning TLBs and Caches Together

*Indian Institute of Science (Master's thesis)*

2019-2020

- Demonstrated the presence of “dead” or unused entries in the L2 TLB
- Designed a dead-page predictor to improve TLB efficiency by bypassing such entries
- Designed a dead-block predictor for last level caches using TLB information
- Showed performance gains of over 8% with low storage overheads

## WORK EXPERIENCE

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### AMD

Bangalore, India

*Senior Silicon Design Engineer (Server Performance Group)* September 2022 - December 2022

- Profiled networking and virtualization workloads on EPYC Bergamo servers using perf / hardware counters, characterizing IO path behavior
- Attributed throughput loss to OS interrupt handling overhead, amplified by Bergamo's core count
- Findings fed into research on utilizing user interrupts (uintr) as a potential kernel bypass path for interrupt delivery

### Nutanix

Bangalore, India

*Member of Technical Staff*

August 2020 - September 2022

- Developed and maintained KVM/QEMU components of the Nutanix AHV hypervisor
- Built VMM live-upgrade support in Libvirt utilizing file-descriptor passing, presented at KVM Forum 2021
- Built a VMM migration monitoring feature, including developing a new daemon to monitor and analyze live migration status

## PUBLICATIONS

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- [1] **HPCA-27:** Chandrashis Mazumdar\*, **Prachatos Mitra\***, Arkaprava Basu, “Dead Page and Dead Block Predictors: Cleaning TLBs and Caches Together”, Proceedings of 27th IEEE International Symposium on High-Performance Computer Architecture (HPCA-27) (virtual), February, 2021. Available at: <https://doi.org/10.1109/HPCA51647.2021.00050>
- [2] **SC-W '23:** **Prachatos Mitra**, Alexandros Daglis, “Filtering Wasteful Vertex Visits in Breadth-First Search”, Proceedings of the SC '23 Workshops of The International Conference on High Performance Computing, Network, Storage, and Analysis, November 2023. Available at: <https://doi.org/10.1145/3624062.3625133>

## TALKS AND PRESENTATIONS

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- [1] **KVM Forum 2021:** **Prachatos Mitra** and Soham Ghosh “Support for Fast and Reliable VMM Live Upgrades in Libvirt”, KVM Forum 2021 (virtual). Available at: [https://www.youtube.com/watch?v=0C\\_0tcIJgpU](https://www.youtube.com/watch?v=0C_0tcIJgpU)

## SERVICE

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- [1] ASPLOS 2027: Artifact Evaluation Committee member

## TECHNICAL SKILLS

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|-------------------------|------------------------------------------------------------------|
| <b>Languages</b>        | C/C++, Python                                                    |
| <b>Performance</b>      | perf, PAPI, hardware counters, roofline analysis                 |
| <b>Simulation tools</b> | SST, gem5, Sniper                                                |
| <b>Systems</b>          | Linux, Libvirt, QEMU, KVM, Bash, L <sup>A</sup> T <sub>E</sub> X |

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\*Equal contribution